Assignment 3

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2026-11-04

```
library(tidyverse)
library(pastecs)

# Load Data
dat <- read.csv('TextMessages.csv')
str(dat)
```

```
## 'data.frame':    50 obs. of  4 variables:
##  $ Group      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Baseline   : int  52 68 85 47 73 57 63 50 66 60 ...
##  $ Six_months : int  32 48 62 16 63 53 59 58 59 57 ...
##  $ Participant: int  1 2 3 4 5 6 7 8 9 10 ...
```

```
## The dataset has 4 variables, all classed as integers.
## It appears that the the variable "Group" is a factor so this will need to be
## updated.
## There is a variable (Participant) indexing the participants
## The other two variables (Baseline, Six_months) are scores taken for each
## participant at different time periods.

## ---- Factorize variables ----

dat$Participant <- as.factor(dat$Participant)
dat$Group <- as.factor(dat$Group)

## ---- Transform data ----
# For ease of visualization, we transform the data to a long format

dat.long <- dat %>%
  pivot_longer(
    cols = c(Baseline, Six_months),
    names_to = 'Time',
    values_to = 'TextMessages'
  )
```

Summary Statistics

```
by(dat.long$TextMessages,
   list (dat.long$Group, dat.long$Time),
   function(X) round(stat.desc(X), 2))
```

```
## : 1
```

```
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##        25.00         0.00         0.00        47.00        85.00        38.00
##          sum       median         mean      SE.mean CI.mean.0.95          var
##      1621.00        64.00        64.84         2.14         4.41       114.06
##      std.dev     coef.var
##        10.68         0.16
## ------------------------------------------------------------
## : 2
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##        25.00         0.00         0.00        46.00        89.00        43.00
##          sum       median         mean      SE.mean CI.mean.0.95          var
##      1640.00        65.00        65.60         2.17         4.47       117.42
##      std.dev     coef.var
##        10.84         0.17
## ------------------------------------------------------------
## : 1
## : Six_months
##      nbr.val     nbr.null       nbr.na          min          max        range
##        25.00         0.00         0.00         9.00        78.00        69.00
##          sum       median         mean      SE.mean CI.mean.0.95          var
##      1324.00        58.00        52.96         3.27         6.74       266.71
##      std.dev     coef.var
##        16.33         0.31
## ------------------------------------------------------------
## : 2
## : Six_months
##      nbr.val     nbr.null       nbr.na          min          max        range
##        25.00         0.00         0.00        46.00        79.00        33.00
##          sum       median         mean      SE.mean CI.mean.0.95          var
##      1546.00        62.00        61.84         1.88         3.88        88.56
##      std.dev     coef.var
##         9.41         0.15
```

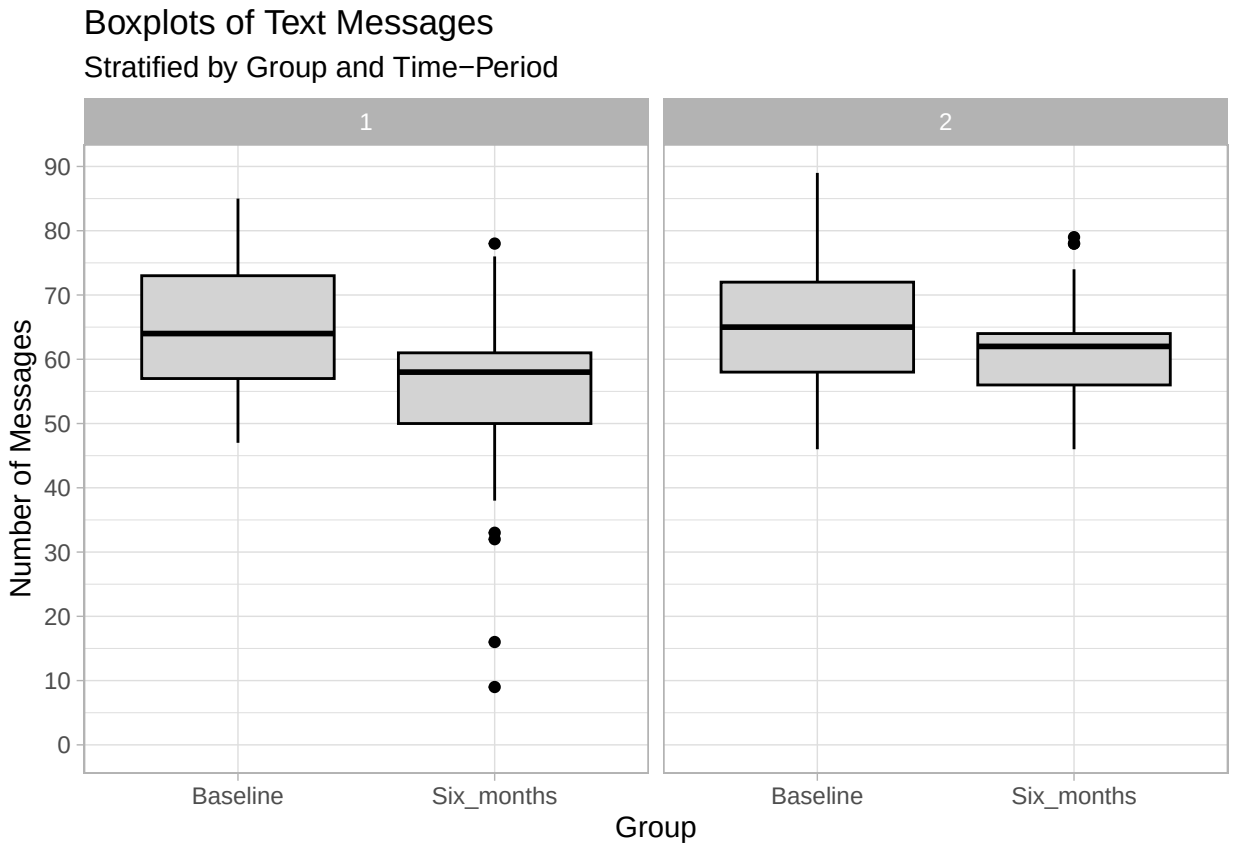
Our summary statistics show a similar average number of texts for both groups at baseline, with Group 1 averaging 64.84 and Group 2 averaging 65.60. After six months, we see a decrease in the means of both groups, however Group 1 had a greater reduction in average texts with 52.96 texts sent on average, while Group 2 had an average of 61.84 texts sent by month six. Group 1 also showed greater variability in the number of texts sent after six months, with its standard deviation increasing from 10.68 to 16.33, while Group 2 showed a slight decrease from 10.84 to 9.41.

Stratified Boxplot

```
# We want to make a boxplots stratified by both Group and Time

ggplot(dat.long, aes(Time, TextMessages)) +
  geom_boxplot(color = 'black', fill = 'lightgrey') +
  # geom_jitter(color = 'blue', width = 0.2) +
  theme_light() +
  coord_cartesian(ylim = c(0, max(dat.long$TextMessages))) +
  scale_y_continuous(breaks = seq(0, max(dat.long$TextMessages)*1.2, by = 10)) +
  labs(
```

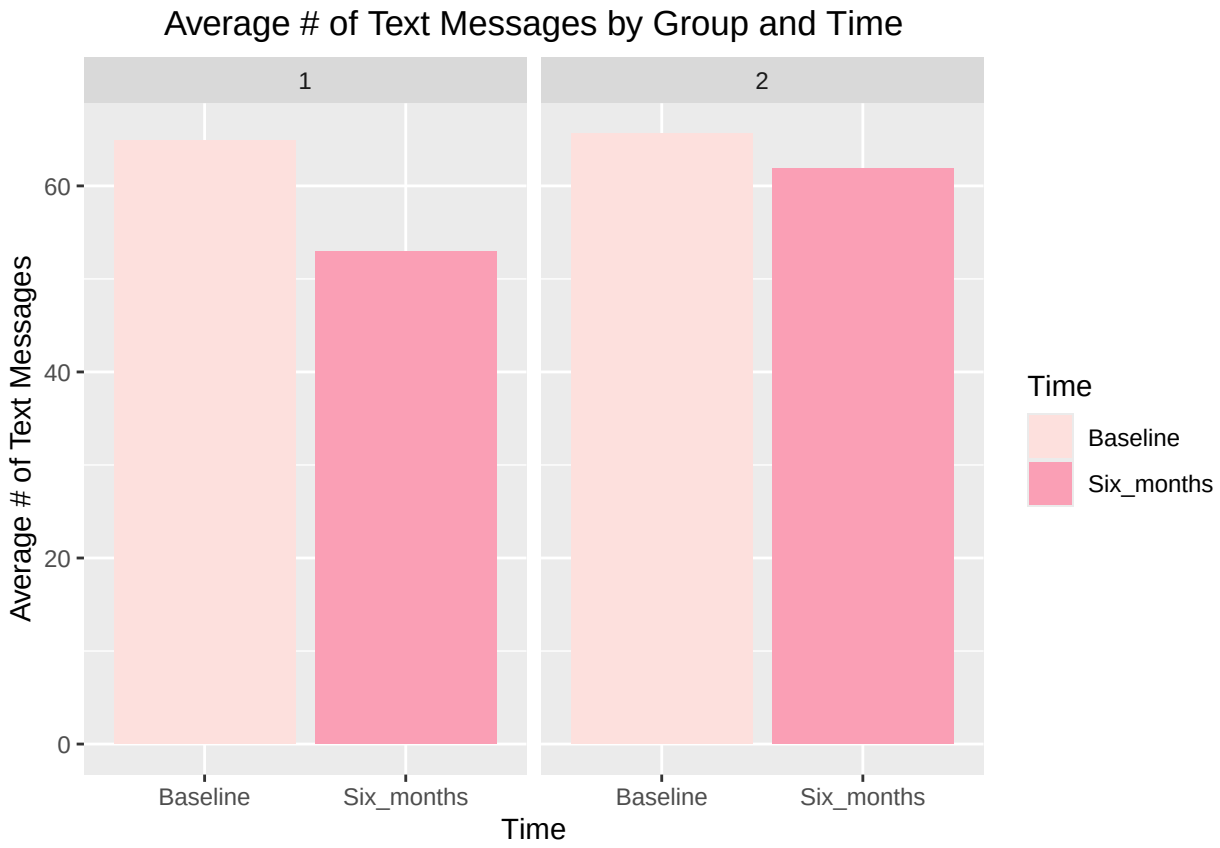
```
    title = 'Boxplots of Text Messages',
    subtitle = 'Stratified by Group and Time-Period',
    x = 'Group',
    y = 'Number of Messages'
  ) +
  facet_wrap(~Group)+
  NULL
```



At baseline, the two groups appear normally distributed and are very similar to each other, with a median around 65. At 6 months, both groups had a decrease in messages; Group 1 appears to have had more of a decrease, and has several outliers that are very low.

Stratified Bar Chart

```
ggplot(dat.long, aes(x=Time, y=TextMessages, fill=Time)) +
  facet_wrap(~Group) +
  stat_summary(fun = mean, geom = "bar") +
  scale_fill_brewer(palette = "RdPu") +
  labs(title = "Average # of Text Messages by Group and Time",
       x = "Time", y = "Average # of Text Messages") +
  theme(plot.title = element_text(hjust = 0.5))
```



The faceted bar chart gives us a clearer view of the decrease in average text messages over time. By separating the data by the groups, we can easily compare the changes from baseline to six months and see that Group 1 experienced a greater decrease than Group 2.